

Quantum Computation and Information

Lecture 5: Quantum Machine Learning

- quantum algorithms for classical ML
 - system of linear equations
 - PCA
 - SVM
- machine learning of quantum data

encoding classical data

- basis encoding: encode data as computational basis of qubit systems
 - examples: binary variable $x \in \{0, 1\}$ as $|x\rangle$, 5 as $|101\rangle$
 - easy preparation, but may require large number of qubits
- amplitude encoding: encode a vector as probability amplitudes of quantum state
 - examples: unit vector b as $\sum_i b_i |i\rangle$
 - small number of qubits, but may be difficult to prepare and read out
- Hamiltonian encoding
 - encode as (parameters of) a Hamiltonian of Ising model
 - * examples: cost Hamiltonian $H_c = \frac{1}{4} \sum_{i,j} w_{ij} (I - Z_i Z_j)$
 - * easy implementation, but limited representation capacity
 - encode matrix as Hamiltonian and Hamiltonian simulation
 - * examples: QAOA algorithm, HHL algorithm
 - * natural for matrices, but require strong conditions for efficiency

system of linear equations

given $A \in \mathbb{R}^{N \times N}$ and $b \in \mathbb{R}^N$, find x satisfying $Ax = b$

- a fundamental problem in sciences and engineering
- classical algorithms
 - $O(N^3)$ for Gaussian elimination
 - $O(Ns\sqrt{k} \log \frac{1}{\epsilon})$ for conjugate gradient method
 - * s maximum number of nonzero entries per row
 - * k condition number of A
 - * ϵ given accuracy

system of linear equations

quantization

- assume $\|b\|_2 = 1$; a unitary that prepares quantum state $|b\rangle = \sum_i b_i |i\rangle$ encoding b in its probability amplitudes (amplitude encoding)
- A is Hermitian and invertible
 - if A is not self-adjoint, solve $\begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} y = \begin{pmatrix} b \\ 0 \end{pmatrix}$ to get $\begin{pmatrix} 0 \\ x \end{pmatrix}$
 - denote by λ_i and $|\psi_i\rangle$ its eigenvalues and eigenvectors
 - A^{-1} and e^{-iAt} have the same set of eigenvectors with eigenvalues $1/\lambda_i$ and $e^{-i\lambda_i t}$, respectively
 - $b = \sum_j \beta_j |\psi_j\rangle$ in terms of these eigenvectors (not explicitly known)

- find a quantum state $|x\rangle \sim \sum_j \frac{\beta_j}{\lambda_j} |\psi_j\rangle$ that approximately encodes solution x^* to $Ax = b$; i.e., for given ϵ

$$\| |x\rangle - |x^*\rangle \| \leq \epsilon$$

- quantum computing outputs/prepares a quantum state encoding x (computation is a physical process!)
- do not output a N -dimensional vector x directly, which would require at least $O(N)$ complexity
- interested in $\langle x | M | x \rangle$ in many applications
- Harrow-Hassidim-Lloyd (HHL) algorithm (Harrow *et al* '09)
 - work well with s -sparse A with low condition number
 - $O(\log N s^2 k^2 / \epsilon)$ complexity

Harrow-Hassidim-Lloyd (HHL) algorithm

- **state preparation:** prepare a quantum state $|b\rangle = \sum_i b_i |i\rangle$ encoding b in its probability amplitudes (amplitude encoding)
 - non-trivial in general, assume an ‘efficient’ unitary exists to prepare the state
 - Grover and Rudolf '08
 - example with $N = 2^3$

$$|0\rangle \Rightarrow \sqrt{\sum_{i=0}^3 |b_i|^2} |0\rangle + \sqrt{\sum_{i=4}^7 |b_i|^2} |1\rangle$$

$$\sqrt{\sum_{i=0}^3 |b_i|^2} |0\rangle |0\rangle \Rightarrow |0\rangle \left(\sqrt{\sum_{i=0}^1 |b_i|^2} |0\rangle + \sqrt{\sum_{i=2}^3 |b_i|^2} |1\rangle \right)$$

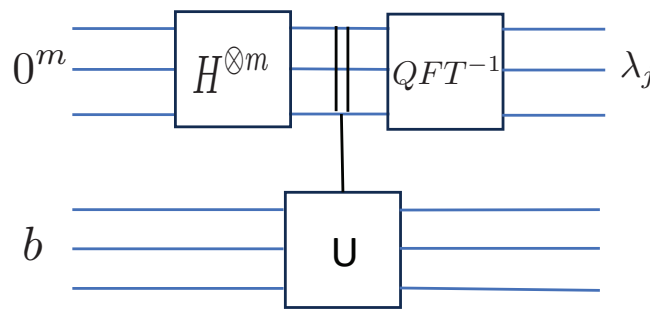
$$\sqrt{\sum_{i=2}^3 |b_i|^2} |01\rangle |0\rangle \Rightarrow |01\rangle (b_2 |0\rangle + b_3 |1\rangle)$$

Harrow-Hassidim-Lloyd (HHL) algorithm

- **Hamiltonian simulation:** transfer Hermitian matrix A into unitary e^{-iAt}
 - find an approximation U such that $\|U - e^{-iAt}\| \leq \epsilon$ with the required accuracy ϵ
 - Trotter formula for additive Hamiltonian $A = \sum_j A_j$:
 $U = (\prod_j e^{-iA_j t/n})^n$ for large n
 - finite Taylor expansion: $U = \sum_{j=0}^n \frac{(-iAt)^j}{j!}$ for sufficiently large n
 - more involved algorithms: quantum walk (Berry *et al* '07), quantum signal processing (Low and Issac '17)
 - require strong conditions for efficient simulations
 - * A is s -sparse, i.e., maximum number of nonzero entries per row is s

Harrow-Hassidim-Lloyd (HHL) algorithm

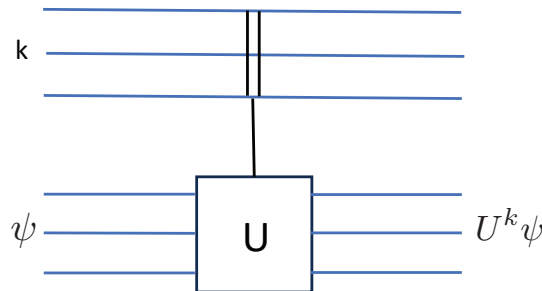
- quantum phase estimation: take $\sum_j \beta_j |\psi_j\rangle |0^{\otimes m}\rangle$ to $\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle$



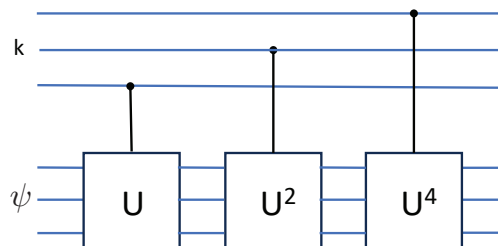
- decompose b in terms of eigenvectors of A and entangle them with states of control/eigenvalue register
- $|\Psi\rangle = \frac{2}{2^m} \sum_{\tau=0}^{2^m-1} \sin \frac{\pi(\tau+1/2)}{2^m} |\tau\rangle$ instead of uniform superposition of all m -qubit states is used to minimize error
- $U = e^{iAt_0/2^m}$ with $t_0 = O(k/\epsilon)$
- obtain an approximation $\tilde{\lambda}_j$ of λ_j with large probability

phase estimation review

- circuit $\Lambda_m(U) : |k\rangle |\psi\rangle \rightarrow |k\rangle U^k |\psi\rangle$



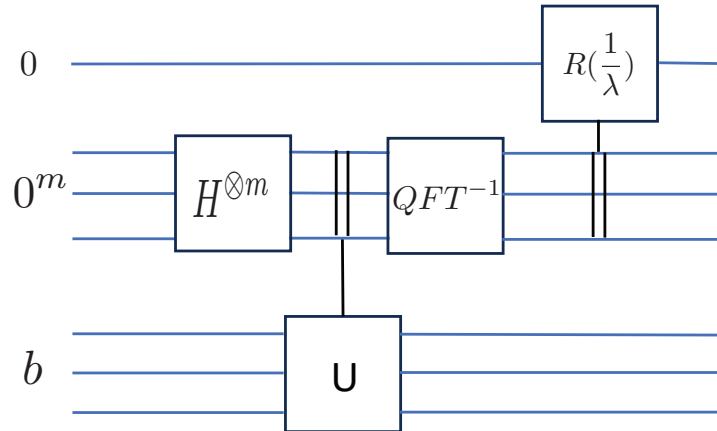
- $k \in \{0, 1, \dots, M-1\}$ with $M = 2^m$; m -qubit states indexed by k
- implement using control- U , control- U^2 , ..., control- $U^{2^{m-1}}$ gates



- estimating θ within m -bit of accuracy is to estimate integer $j < M$ such that $\frac{j}{M}$ is the closest approximation to θ

Harrow-Hassidim-Lloyd (HHL) algorithm

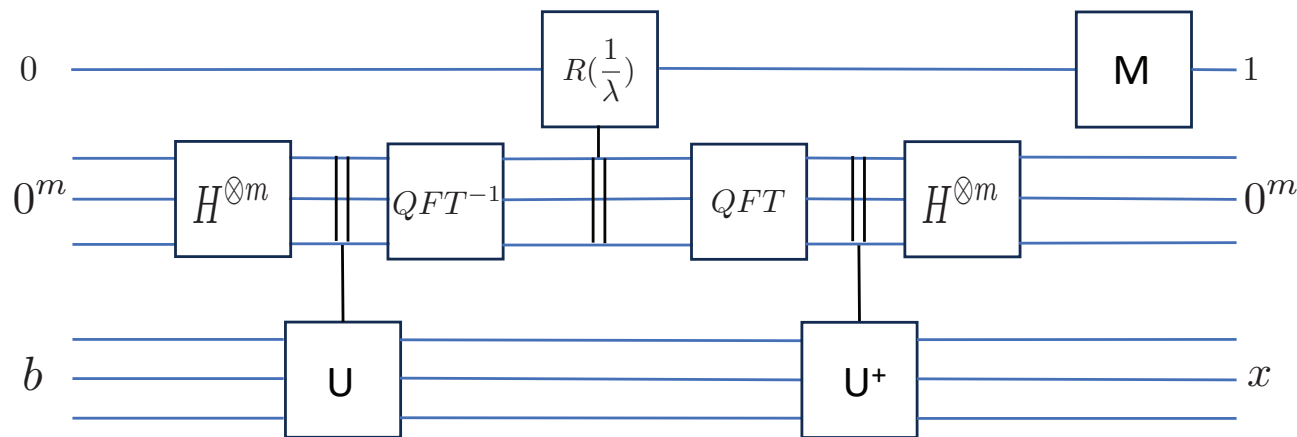
- rotation of ancillary qubit conditioned on $|\lambda_j\rangle$: take $\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle |0\rangle$ to $\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle (\sqrt{1 - (\frac{C}{\lambda_j})^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle)$



- $C = O(1/k)$
- more nuances here

Harrow-Hassidim-Lloyd (HHL) algorithm

- **uncompute $|\lambda_j\rangle$** : reverse quantum phase estimation to take $\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle (\sqrt{1 - (\frac{C}{\lambda_j})^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle)$ to $\sum_j \beta_j |\psi_j\rangle |0^m\rangle (\sqrt{1 - (\frac{C}{\lambda_j})^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle)$



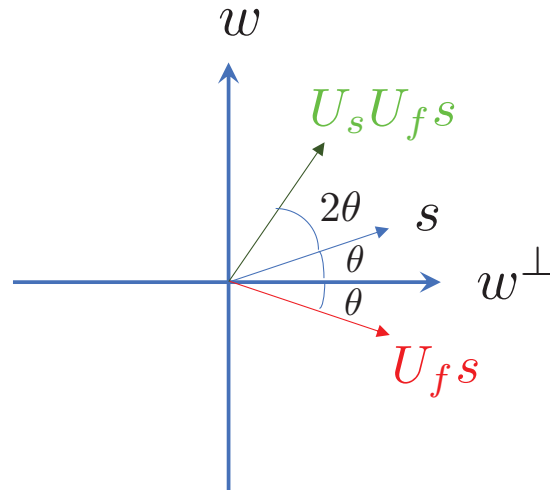
- **conditioned on measurement result of 1 (rejection sampling)**: obtain state $\sim \sum_j \frac{C\beta_j}{\lambda_j} |\psi_j\rangle$, corresponding to $|x\rangle = \sum_j \frac{\beta_j}{\lambda_j} |\psi_j\rangle$ up to a normalization

Harrow-Hassidim-Lloyd (HHL) algorithm

- **amplitude amplification:** achieve quadratic speedup in number of runs
 - suppose $U |0^n\rangle |0\rangle = \cos \theta |\Psi_0\rangle |0\rangle + \sin \theta |\Psi_1\rangle |1\rangle$ for given unitary U . define $S_1 = I_m \otimes (I - 2|1\rangle\langle 1|)$, $S_i = I_{m+1} - 2|0^m\rangle|0\rangle\langle 0|\langle 0^m|$ and $Q = -US_iU^\dagger S_1$, then

$$Q^k U |0^n\rangle |0\rangle = \cos (2k + 1)\theta |\Psi_0\rangle |0\rangle + \sin (2k + 1)\theta |\Psi_1\rangle |1\rangle$$

- generalization of Grover iteration



review on HHL algorithm

given a Hermitian and invertible matrix $A \in \mathbb{R}^{N \times N}$ and a unit vector $b \in \mathbb{R}^N$, find x satisfying $Ax = b$

- denote by λ_j and v_j eigenvalues and eigenvectors of A
- A^{-1} and e^{-iAt} have the same set of eigenvectors with eigenvalues $1/\lambda_j$ and $e^{-i\lambda_j t}$, respectively
- $x = A^{-1}b = A^{-1}(\sum_j \beta_j v_j) = \sum_j \beta_j A^{-1}v_j = \sum_j \frac{\beta_j}{\lambda_j} v_j$

quantization

- $b = \sum_j \beta_j v_j \Rightarrow |b\rangle = \sum_j \beta_j |\psi_j\rangle$, compute $|x\rangle \sim \sum_j \frac{\beta_j}{\lambda_j} |\psi_j\rangle$

key steps

- exponentiation e^{-iAt} of Hermitian matrix and Hamilton simulation (do not get into details)
 - an important building block for many quantum ML algorithms
 - efficient simulation often relies on specific structures such as sparsity
- quantum phase estimation to decompose states of input-eigenvalue registers into eigenstates and eigenvalues

$$|b\rangle |0^{\otimes m}\rangle \Rightarrow \sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle$$

- controlled rotation of an ancillary qubit conditioned on λ_j

$$\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle |0\rangle \Rightarrow \sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle \left(\sqrt{1 - \left(\frac{C}{\lambda_j}\right)^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right)$$

- uncompute (reverse quantum phase estimation) to disentangle eigenvalue register

$$\sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle (\sqrt{1 - (\frac{C}{\lambda_j})^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle) \Rightarrow \sum_j \beta_j |\psi_j\rangle |0^m\rangle (\sqrt{1 - (\frac{C}{\lambda_j})^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle)$$

- conditioned on measurement outcome 1 of ancillary qubit (rejection sampling)

$$\sum_j \frac{C\beta_j}{\lambda_j} |\psi_j\rangle \sim |x\rangle$$

- amplitude amplification to reduce complexity

- given $U |0^{\otimes n}\rangle |0\rangle = \cos \theta |\Psi_0\rangle |0\rangle + \sin \theta |\Psi_1\rangle |1\rangle$
- define $S_1 = I_n \otimes (I - 2 |1\rangle \langle 1|)$, $S_i = I_{n+1} - 2 |0^{\otimes n}\rangle |0\rangle \langle 0| \langle 0^{\otimes n}|$
- define $Q = -US_iU^\dagger S_1$, then

$$Q^k U |0^{\otimes n}\rangle |0\rangle = \cos (2k + 1)\theta |\Psi_0\rangle |0\rangle + \sin (2k + 1)\theta |\Psi_1\rangle |1\rangle$$

- quadratic speedup

basic linear algebra operations

- quantum Fourier transformation (QFT)
- finding eigenvalues and eigenstates by quantum phase estimation (QPE)

$$|b\rangle \left|0^{\otimes n}\right\rangle \xrightarrow{\text{QPE}} \sum_j \beta_i |\psi_j\rangle |\lambda_j\rangle$$

- matrix inversion (solving linear equations $Ax = b$)

$$|b\rangle \left|0^{\otimes m}\right\rangle |0\rangle \xrightarrow{\text{QPE}} \sum_j \beta_i |\psi_j\rangle |\lambda_j\rangle |0\rangle$$

$$\begin{aligned} & \xrightarrow{R(1/\lambda_j)} \sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle \left(\sqrt{1 - \left(\frac{C}{\lambda_j}\right)^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right) \\ & \xrightarrow{\text{QFE}^{-1}} \sum_j \beta_i |\psi_j\rangle \left|0^{\otimes m}\right\rangle \left(\sqrt{1 - \left(\frac{C}{\lambda_j}\right)^2} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right) \xrightarrow[\text{on outcome 1}]{\text{conditioned}} \sum_j \frac{C\beta_i}{\lambda_j} |\psi_j\rangle \end{aligned}$$

- matrix multiplication ($x = Ab$)

$$\begin{aligned}
 & |b\rangle |0^{\otimes m}\rangle |0\rangle \xrightarrow{\text{QPE}} \sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle |0\rangle \\
 & \xrightarrow{R(\lambda_j)} \sum_j \beta_j |\psi_j\rangle |\lambda_j\rangle (\sqrt{1 - (C\lambda_j)^2} |0\rangle + C\lambda_j |1\rangle) \\
 & \xrightarrow{\text{QFE}^{-1}} \sum_j \beta_j |\psi_j\rangle |0^{\otimes m}\rangle (\sqrt{1 - (C\lambda_j)^2} |0\rangle + C\lambda_j |1\rangle) \xrightarrow[\text{on outcome 1}]{\text{conditioned}} \sum_j C\beta_j \lambda_j |\psi_j\rangle
 \end{aligned}$$

- exponential quantum speedup ($O((\log N)^{O(1)})$ instead of $O(N^{O(1)})$)
 - exponential representation capacity (2^n -D space for n -qubit)
 - quantum parallel (simultaneous operation on all basis states of superposition)
- can do lots of machine learning and optimization using these basic linear algebra operations
 - quantum accelerator?

least-squares data fitting

given data $\{x_i, y_i\}_{i=1, 2, \dots, m}$ with $x_i \in \mathbb{R}^n$, $y_i \in \mathbb{R}$, find linear model $y = k^T x$ so as to minimize $\sum_i (y_i - k^T x_i)^2$

$$\min_k \|Y - Xk\|_2^2$$

- $Y = \{y_1, y_2, \dots, y_m\}^T$ and $X = \{x_1^T, x_2^T, \dots, x_m^T\}^T$
- solution

$$k = (X^T X)^{-1} X^T Y$$

- how to solve it “quantumly”?

- use HHL algorithm
 - define $A = \begin{pmatrix} 0 & X^T \\ X & 0 \end{pmatrix}$ and $\tilde{A} = \begin{pmatrix} X^T X & 0 \\ 0 & X X^T \end{pmatrix}$
 - apply HHL to $\tilde{A}^{-1} A (\mathbf{0}^T, Y^T)^T$
- can solve unconstrained and equality-constrained quadratic programming
- can potentially be use for interior-point method for solving any convex optimization problems with inequality constraints